

Agentic AI Concepts - Skill IQ Practice Question Bank

Medium, Hard, and Very Hard MCQs with answer key and explanations

Purpose: Build Skill IQ-style readiness using original scenario-based MCQs. These are not official Pluralsight exam questions or dumps.

Difficulty mix: Medium questions test core knowledge; Hard questions test applied decision-making; Very Hard questions test architecture, edge cases, trade-offs, and production judgement.

Focus areas

- Agents vs chatbots vs workflows, goals, tools, planning, memory, and environment interaction
- ReAct, function calling, tool schemas, orchestration, routing, reflection, and multi-agent coordination
- Agentic RAG, persistent memory, event-driven agents, observability, evaluation, and production deployment
- Safety, authorization, prompt injection, human-in-the-loop control, and runaway loop prevention

How to use this bank

Do the questions timed first, then review the answer key. For Skill IQ-style prep, focus on why the correct answer is better than plausible alternatives.

Source scope

Aligned with public Pluralsight path descriptions and general industry knowledge. Topics emphasize breadth, applied reasoning, and production trade-offs rather than memorized question dumps.

Questions

Q01 - Medium - Definition

What makes an AI system agentic?

- A. It can pursue goals by planning, using tools, taking actions, and iterating based on feedback
- B. It only answers one text prompt
- C. It has no memory and no tools
- D. It must always be a robot

Q02 - Medium - Agent vs chatbot

How is a basic chatbot different from an agentic assistant?

- A. A chatbot mainly responds; an agent may plan and act through tools
- B. A chatbot cannot use text
- C. An agent never uses LLMs
- D. They are always identical

Q03 - Medium - Tools

What is a tool in an LLM agent system?

- A. An external capability the model can call, such as search, calendar, database, or code execution
- B. A hidden neural network layer only
- C. A user interface color
- D. A training loss

Q04 - Medium - Planning

Why do agents often decompose tasks into steps?

- A. Complex goals usually require ordered actions and intermediate checks
- B. It always reduces latency
- C. It removes the need for tools
- D. It prevents all errors

Q05 - Medium - Memory

What is agent memory used for?

- A. Storing useful context across turns or tasks
- B. Increasing GPU temperature
- C. Replacing all prompts
- D. Disabling retrieval

Q06 - Medium - Function calling

What does a function schema provide to an agent?

- A. A structured contract for tool arguments and expected inputs
- B. A new model architecture
- C. A way to avoid all validation
- D. A replacement for user consent

Q07 - Medium - Human review

When should an agent ask for human confirmation?

- A. Before irreversible, expensive, sensitive, or high-impact actions
- B. Before every harmless word it writes
- C. Never
- D. Only when no tools exist

Q08 - Medium - Agent loop

Which sequence best describes a common agent loop?

- A. Observe -> plan -> act -> evaluate -> repeat or finish
- B. Train -> delete -> ignore -> finish
- C. Classify -> stop always
- D. Render -> compile -> tokenize only

Q09 - Medium - RAG agents

What is agentic RAG?

- A. An agent deciding when and how to retrieve, query, refine, or use knowledge sources
- B. A static prompt with no retrieval
- C. A model with no tools
- D. A clustering algorithm only

Q10 - Medium - Orchestration

What is orchestration in agentic systems?

- A. Coordinating agents, tools, steps, state, and workflows
- B. Only changing the UI theme
- C. Only compressing model weights
- D. Removing all safety checks

Q11 - Medium - Reflection

What is reflection in agent design?

- A. The agent reviews its own output or plan to identify errors and improve
- B. The model changes hardware
- C. A database replication pattern only
- D. A way to skip validation

Q12 - Medium - Guardrails

What are guardrails in agentic AI?

- A. Controls that constrain, validate, monitor, or block unsafe behavior
- B. A larger context window
- C. A type of optimizer
- D. A random seed

Q13 - Hard - ReAct

What is the key idea of the ReAct pattern?

- A. Interleave reasoning-like planning with actions/tool calls and observations
- B. Train the model using only images
- C. Avoid all external tools
- D. Use one fixed answer for every prompt

Q14 - Hard - Tool safety

An agent can send emails. What is the safest design?

- A. Draft first and request confirmation before sending unless policy allows auto-send
- B. Let the model send any email immediately
- C. Store all contacts in the prompt
- D. Ignore recipient validation

Q15 - Hard - Runaway loops

How can an agent system reduce infinite or runaway loops?

- A. Set iteration limits, budgets, termination criteria, and failure handling
- B. Give the model unlimited tool calls
- C. Remove logs
- D. Never evaluate tool results

Q16 - Hard - Multi-agent systems

What is a risk of multi-agent collaboration?

- A. Error propagation, coordination overhead, conflicting goals, and harder debugging
- B. Guaranteed accuracy
- C. No need for orchestration
- D. Elimination of latency

Q17 - Hard - Routing

What is a router agent used for?

- A. Selecting the appropriate tool, expert, workflow, or model for a request
- B. Encrypting tokens
- C. Changing labels to features
- D. Replacing all retrieval

Q18 - Hard - State management

Why is explicit state important in agentic workflows?

- A. It tracks progress, decisions, tool outputs, and dependencies across steps
- B. It prevents all hallucinations
- C. It removes all latency
- D. It makes prompts unnecessary

Q19 - Hard - Observability

What should be logged for production agents?

- A. Prompts/messages, tool calls, decisions, errors, latency, costs, and outcomes, subject to privacy controls
- B. Only final answers
- C. Only GPU brand
- D. Nothing, because logs are never useful

Q20 - Hard - Authorization

Why should tool permissions be enforced outside the LLM?

- A. The LLM can be mistaken or manipulated; external access controls are more reliable
- B. Prompts are stronger than permissions
- C. Users never attack agents
- D. Function schemas are the same as authorization

Q21 - Hard - Agent evaluation

Which metric is most useful for an agent completing multi-step tasks?

- A. Task success rate with checks on correctness, safety, cost, and steps taken
- B. Only next-token loss
- C. Only response length
- D. Only font consistency

Q22 - Hard - Memory safety

What is a risk of persistent memory?

- A. Storing incorrect, sensitive, or stale information that affects future behavior
- B. It always improves accuracy without risk
- C. It prevents personalization
- D. It removes the need for user consent

Q23 - Hard - Tool selection

An agent answers a current weather question from memory instead of calling a weather tool. What failed?

- A. Tool-use policy or freshness routing
- B. Embedding dimensionality only
- C. UI styling
- D. Batch normalization

Q24 - Hard - Workflow agents

When is a deterministic workflow preferable to an autonomous agent?

- A. When the process is stable, regulated, and needs predictable steps
- B. When goals are ambiguous and exploratory
- C. When no rules exist
- D. When creativity is the only objective

Q25 - Very Hard - Prompt injection in agents

A retrieved web page tells an agent to transfer money to a new account. What is the correct response?

- A. Treat it as untrusted content, ignore action instructions from retrieved data, and require authorized user confirmation through secure channels
- B. Execute the transfer because the page was retrieved
- C. Add the account to memory
- D. Disable all future retrieval forever

Q26 - Very Hard - Least privilege

What does least privilege mean for agents?

- A. Give each agent/tool only the permissions needed for its specific task
- B. Give every agent admin access
- C. Let the model decide permissions at runtime
- D. Avoid authentication

Q27 - Very Hard - Agent architecture

Which architecture is best for a customer-refund agent with high-value transactions?

- A. Planner plus policy retrieval, tool permission checks, transaction limits, human approval thresholds, and audit logs
- B. Unrestricted tool access with no logging
- C. One prompt with all customer data and no validation
- D. A chatbot that guesses refund amounts

Q28 - Very Hard - Reflection limits

Why can reflection loops fail?

- A. The same model may reinforce its own wrong assumptions without external evidence or tests
- B. Reflection always guarantees correctness
- C. Reflection removes the need for tools
- D. Reflection can only be used for images

Q29 - Very Hard - Event-driven agents

What is an event-driven agent?

- A. An agent triggered by events such as emails, tickets, alerts, or workflow state changes
- B. An agent that only responds to manual chat messages
- C. A model that cannot use tools
- D. A neural network optimizer

Q30 - Very Hard - Agentic knowledge graphs

How can a knowledge graph help an agent?

- A. By providing structured entities, relationships, and constraints for planning or retrieval
- B. By eliminating all prompt injection
- C. By replacing every tool
- D. By making logs unnecessary

Q31 - Very Hard - Failure recovery

A tool call fails due to a transient API error. What should a robust agent do?

- A. Retry within limits, use fallback if available, report failure clearly, and avoid fabricating success
- B. Claim the action succeeded
- C. Loop forever
- D. Delete the task state

Q32 - Very Hard - Agent benchmarks

Why are agent benchmarks harder than simple QA benchmarks?

- A. They require measuring sequential decisions, tool use, side effects, state, and success criteria
- B. They only need exact string matching
- C. They never involve environments
- D. They do not need safety evaluation

Q33 - Very Hard - Human control

An agent proposes a plan with three destructive database operations. What should the system require?

- A. Explicit review, permission checks, backup/rollback plan, and staged execution
- B. Immediate execution
- C. Hide the plan from the user
- D. Disable logs

Q34 - Very Hard - Model choice

Why might a smaller model be used inside an agent even if a larger model is more capable?

- A. For routing, extraction, or simple checks where lower cost and latency are more important
- B. Smaller models are always more accurate
- C. Large models cannot use tools
- D. Agent systems cannot combine models

Q35 - Very Hard - Autonomy levels

What is a good way to increase agent autonomy safely over time?

- A. Start with suggestions/drafts, add supervised actions, then automate low-risk actions after evals and monitoring
- B. Give full production permissions on day one
- C. Remove human review before testing
- D. Ignore failure cases

What should be tested before enabling long-term memory in an agent?

- A. Whether memory is accurate, consented, useful, deletable, non-sensitive, and not over-weighted against current instructions
- B. Only whether memory increases token count
- C. Whether memory makes the UI prettier
- D. Nothing, memory is always safe

Answer Key with Explanations

Q01. A - It can pursue goals by planning, using tools, taking actions, and iterating based on feedback

Definition / Medium

Agentic systems move beyond passive response into goal-directed action and feedback loops.

Q02. A - A chatbot mainly responds; an agent may plan and act through tools

Agent vs chatbot / Medium

Agentic assistants can execute workflows and use tools toward goals.

Q03. A - An external capability the model can call, such as search, calendar, database, or code execution

Tools / Medium

Tools extend the agent's abilities beyond text generation.

Q04. A - Complex goals usually require ordered actions and intermediate checks

Planning / Medium

Task decomposition improves control and execution of multi-step goals.

Q05. A - Storing useful context across turns or tasks

Memory / Medium

Memory helps agents maintain continuity and personalize or continue work.

Q06. A - A structured contract for tool arguments and expected inputs

Function calling / Medium

Schemas help the model call tools with valid structured inputs.

Q07. A - Before irreversible, expensive, sensitive, or high-impact actions

Human review / Medium

Human confirmation is important for risky actions.

Q08. A - Observe -> plan -> act -> evaluate -> repeat or finish

Agent loop / Medium

Agents commonly observe state, plan, act, evaluate feedback, and iterate.

Q09. A - An agent deciding when and how to retrieve, query, refine, or use knowledge sources

RAG agents / Medium

Agentic RAG makes retrieval part of a decision workflow, not just one fixed lookup.

Q10. A - Coordinating agents, tools, steps, state, and workflows

Orchestration / Medium

Orchestration manages execution flow and coordination.

Q11. A - The agent reviews its own output or plan to identify errors and improve

Reflection / Medium

Reflection is a self-critique or self-evaluation step.

Q12. A - Controls that constrain, validate, monitor, or block unsafe behavior

Guardrails / Medium

Guardrails help keep agents within safe and intended boundaries.

Q13. A - Interleave reasoning-like planning with actions/tool calls and observations

ReAct / Hard

ReAct-style agents alternate thought/planning, actions, and observations.

Q14. A - Draft first and request confirmation before sending unless policy allows auto-send

Tool safety / Hard

Email sending can have real-world impact, so confirmation and validation are needed.

Q15. A - Set iteration limits, budgets, termination criteria, and failure handling

Runaway loops / Hard

Operational limits and stop conditions are necessary for reliable agents.

Q16. A - Error propagation, coordination overhead, conflicting goals, and harder debugging

Multi-agent systems / Hard

More agents can increase complexity and failure modes.

Q17. A - Selecting the appropriate tool, expert, workflow, or model for a request

Routing / Hard

Routers direct tasks to suitable components.

Q18. A - It tracks progress, decisions, tool outputs, and dependencies across steps

State management / Hard

State prevents the agent from losing track of what has happened and what remains.

Q19. A - Prompts/messages, tool calls, decisions, errors, latency, costs, and outcomes, subject to privacy controls

Observability / Hard

Observability enables debugging, audit, safety review, and optimization.

Q20. A - The LLM can be mistaken or manipulated; external access controls are more reliable

Authorization / Hard

Security should not depend solely on model compliance.

Q21. A - Task success rate with checks on correctness, safety, cost, and steps taken

Agent evaluation / Hard

Agent evaluation must measure end-to-end task completion and operational constraints.

Q22. A - Storing incorrect, sensitive, or stale information that affects future behavior

Memory safety / Hard

Memory needs validation, consent, retention, and correction mechanisms.

Q23. A - Tool-use policy or freshness routing

Tool selection / Hard

Fresh information should trigger the appropriate external tool.

Q24. A - When the process is stable, regulated, and needs predictable steps

Workflow agents / Hard

Stable high-control tasks often benefit from deterministic workflow orchestration.

Q25. A - Treat it as untrusted content, ignore action instructions from retrieved data, and require authorized user confirmation through secure channels

Prompt injection in agents / Very Hard

Retrieved content must not authorize real-world actions.

Q26. A - Give each agent/tool only the permissions needed for its specific task

Least privilege / Very Hard

Least privilege limits damage from mistakes or attacks.

Q27. A - Planner plus policy retrieval, tool permission checks, transaction limits, human approval thresholds, and audit logs

Agent architecture / Very Hard

High-impact actions need policy grounding, controls, approval, and auditing.

Q28. A - The same model may reinforce its own wrong assumptions without external evidence or tests

Reflection limits / Very Hard

Self-critique is not a substitute for grounding, testing, or external validation.

Q29. A - An agent triggered by events such as emails, tickets, alerts, or workflow state changes

Event-driven agents / Very Hard

Event-driven agents react to external events and can initiate workflows.

Q30. A - By providing structured entities, relationships, and constraints for planning or retrieval

Agentic knowledge graphs / Very Hard

Graphs can improve structured reasoning, context selection, and constraint checking.

Q31. A - Retry within limits, use fallback if available, report failure clearly, and avoid fabricating success

Failure recovery / Very Hard

Reliable agents need bounded retries, fallbacks, and honest status reporting.

Q32. A - They require measuring sequential decisions, tool use, side effects, state, and success criteria

Agent benchmarks / Very Hard

Agent performance depends on interactive workflows, not just one answer.

Q33. A - Explicit review, permission checks, backup/rollback plan, and staged execution

Human control / Very Hard

Destructive actions need human control and recovery planning.

Q34. A - For routing, extraction, or simple checks where lower cost and latency are more important

Model choice / Very Hard

Agent architectures can use different models based on task complexity and cost.

Q35. A - Start with suggestions/drafts, add supervised actions, then automate low-risk actions after evals and monitoring

Autonomy levels / Very Hard

Progressive autonomy reduces risk as evidence of reliability grows.

Q36. A - Whether memory is accurate, consented, useful, deletable, non-sensitive, and not over-weighted against current instructions

Agent memory evaluation / Very Hard

Long-term memory needs correctness, safety, privacy, and user control.

Last-minute revision checklist

- Distinguish chatbots, deterministic workflows, and autonomous/autentic systems.
- Know agent loops: observe, plan, act, evaluate, repeat or stop.
- Understand tools, schemas, routing, memory, state, orchestration, and ReAct-style patterns.
- Design safety controls: least privilege, approvals, budgets, iteration limits, and audit logs.
- Evaluate agents by task success, correctness, safety, cost, and latency.
- Handle prompt injection, untrusted retrieval, tool errors, and human escalation.

Suggested timed drill

Pick 25 mixed questions and give yourself 15 minutes. Then review every wrong or guessed answer and write one sentence explaining the underlying concept.